

JavaScript CI Activity Analysis – Report

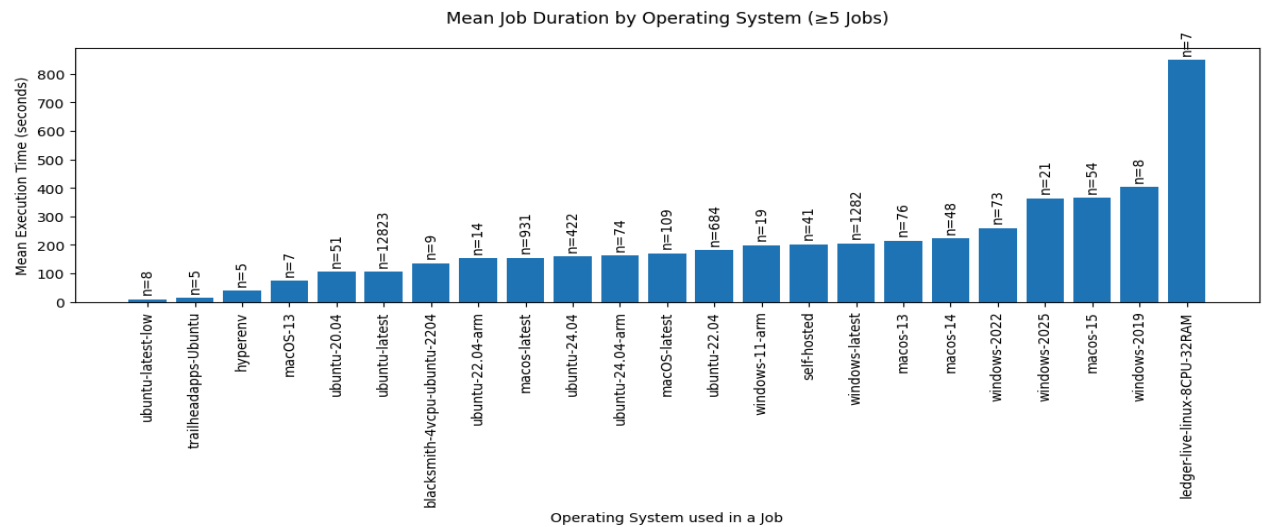
This task analyzes GitHub Actions usage in JavaScript repositories with 10+ stars.

Dataset & Processing:

- Source: [SEART GitHub datadump \(Aug 2024\)](#).
- Filtered repos: 24,612 (JavaScript, >10 commits, active since May 2024).
- Repos with workflows .yml files: 12,215.
- Workflows with runs on github action: 11,067
- Workflows runs without bot: 8311
- Jobs retrieved: 20,528 => cleaned to 16,816.
- Final workflow data: execution times, matrix configs/number of jobs
- Final job data: operating system, number of steps

Observations

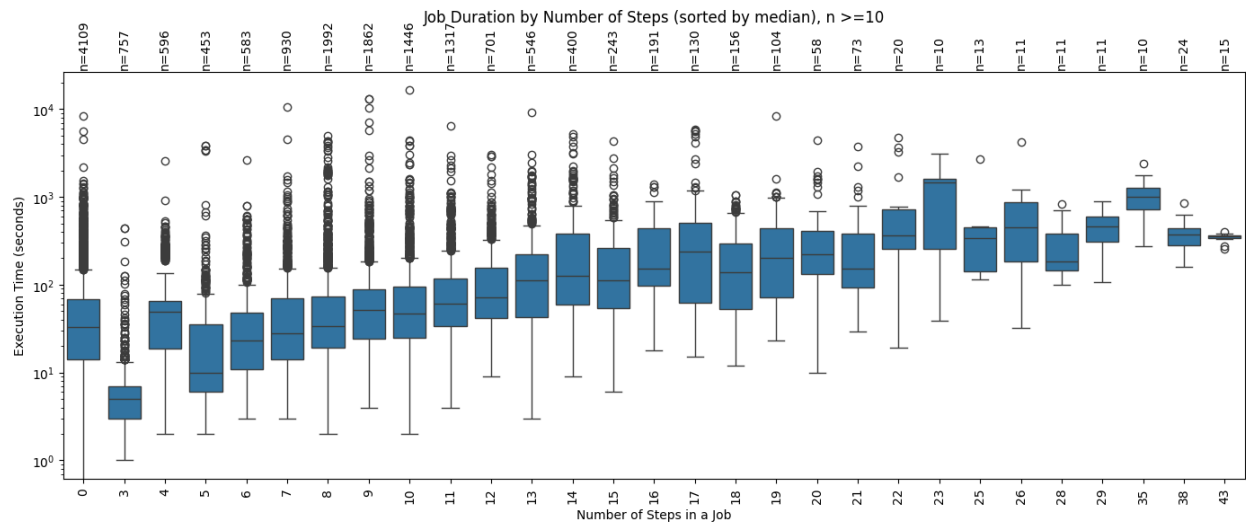
Job Duration by Operating System



Observation:

- Linux-based runners (Ubuntu variants) are the most frequently used (e.g., ubuntu-latest, ubuntu-20.04) and show lower average execution times compared to Windows and macOS.
- Windows runners (e.g., windows-2022, windows-2019) exhibit longer mean job durations, often 2–3× higher than Linux.
- The ledger-live-linux-8CPU-32RAM runner stands out with the highest mean job duration, though its usage count is very small.

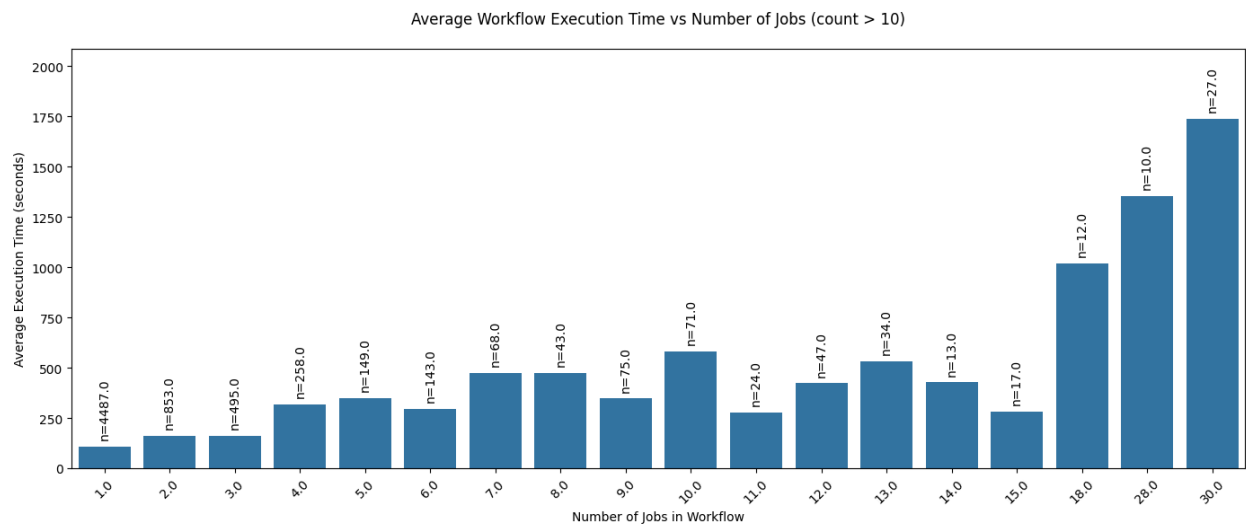
Job Duration by Number of Steps



Observation:

- Jobs with fewer steps (0–5) dominate the dataset (e.g., ≈4,109 jobs with 0 steps), showing relatively low median execution times, mostly under a few hundred seconds.
- As the number of steps increases, the median execution time generally grows, with clear upward shifts around 10–20 steps.
- Spikes at higher step counts (e.g., 23, 29 steps) indicate that some jobs with more steps take significantly longer to complete.
- Extremely high execution times across many step counts suggest the presence of outliers, where certain steps involve complex or resource-heavy tasks that disproportionately increase duration.

Workflow Duration by Number of Jobs



Observation:

- Workflows with 1 job dominate the dataset (≈4,487), showing an average execution time of just over 100 seconds.
- As the number of jobs increases, the average execution time rises overall, with noticeable spikes around 7, 10, 18, and 30 jobs, where workflows take substantially longer to finish.